

ZS-A19

Z-Spin Boundary Tension as Geometric Dust: A Rank-Absorption No-Go and a Closed Boundary-Charge (ZHCS) Program — Stückelberg→BF Dualization, Actual- W_{bc} Connectivity, the Boundary BFV Differential, and Brown–Kuchař Canonical Branching Lifting C1 to DERIVED and C2 to DERIVED-CONDITIONAL on a Single Source

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GitHub: <https://github.com/KennyKang-git/zspin>

Locked inputs: $A = 35/437 \approx 0.080092$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$. Zero new fitted parameters.

Verification: 22/22 independent recomputations PASS (polyhedral, graph-spectral, cross-transfer structure, symplectic branching, and the C1/C2 closure computations; Appendices C–H). v3.0 carries out the two decisive action-level computations the v2.4 review identified: (i) the S10 Stückelberg $U(1)$ dualizes to a 2-form–1-form BF coupling and the boundary BFV differential’s degree-0→1 block equals $q_Z d_\Gamma (\|\Omega \partial \Sigma^{(0 \rightarrow 1)} - q_Z d_\Gamma\| = 0)$; and (ii) the Brown–Kuchař clock-energy momentum $P_T = -H_{ZS}$ canonically branches with $\Theta_c + \Theta_b = \Theta_*$ and node-uniform normalization $\varepsilon_c = \varepsilon_b$. With the actual corpus W_{bc} connecting all six baryon nodes to the connected TI block (no cube rewiring), this lifts C1 to DERIVED and C2 to DERIVED-CONDITIONAL on a single source. A19.NG1, A19.NG2 PROVEN and intact; ZS-A18 NO-GO frozen. Zero new fitted parameters.

§0. Abstract

ZS-A18 (v1.5, frozen) proved that a smooth massless Z-Spin Goldstone gradient cannot be the recombination cold dark matter (CDM) that sustains the third CMB acoustic peak: a smooth massless field has $w \in \{-1/3, +1\}$, never $w = 0$. This paper does not contest that NO-GO; it proposes a different carrier. We abandon the premise that dark matter is a collection of objects (Planck-mass vortex cores, vortons, primordial-black-hole relics) — the premise responsible for the over-closure, $\sim 10^9$ GeV formation-scale, and current-carrier obstructions, all of which are artifacts of the particle ontology — and instead implement ZS-F2’s definition of CDM as a geometric boundary tension through a conserved pullback / Brown–Kuchař dust ($w = 0$, $c_s^2 = 0$, $\rho \propto a^{-3}$) that leaves the Goldstone exactly massless (§3–§4).

The fraction $\Omega_{\text{cdm}} = F(\text{TI})/Q^2 = 32/121$ is read as an equivariant projection-class rank: the truncated-icosahedron face module embeds A_5 -equivariantly in the $Q^2 = 121$ channel algebra as a rank-32 projector with canonical normalized trace 32/121 (Theorem A19.2), the “32 charge” being neither a spatial-topological charge (Lemma A19.3a NO-GO) nor a metric tension (Lemma A19.3b NO-GO). Two proven obstructions discipline the program: **Theorem A19.NG1** (the unique faithful unital embedding $A_{ZS} = M_3 \oplus \mathbb{C} \oplus M_5 \rightarrow M_{11}$ forces center weights $(3,3,5)/11 \neq$ the ZS-F23 $(3,2,6)/11$, so BT-C and ZS-F23 Condition C are distinct traces) and **Theorem A19.NG2** (for a scalar clock the seam projector

enters the bulk action only through $\text{Tr } P_c = 32$, removed by $\lambda_c \rightarrow \lambda_c/32$, so no bulk rank-lock fixes the cold fraction).

A19.NG2 forces the rank out of the bulk and onto a conserved boundary canonical charge: the **ZHCS** program puts one harmonic parent charge p on a 38-node seam graph of the 32 cold (truncated-icosahedron) faces and the 6 baryon (cube) faces, so that $Q_c : Q_b = 32 : 6$ is the rescaling-invariant ratio of two projections of one parent charge (never written into the action). v2.0 promoted this to DERIVED-CONDITIONAL on a single operator identification; v2.1–v2.3 corrected the substrate and consolidated the gaps. v2.4 makes the reduction honest and the conditions checkable. A corrected recomputation shows (i) the truncated-octahedron incidence kernel is **two-dimensional**, so v2.0's "unique tO null mode" cross-support argument is **retracted**; (ii) the six tO squares are **mutually non-adjacent**; and (iii) the BV-BFV mQME gives $\Omega\partial\Sigma\psi = -\hbar^2\Delta V\psi$, not $\Omega\partial\Sigma\psi = 0$. The cross-block claim is then re-grounded **structurally**: the corpus fixes the X–Z–Y mediation as the rank-1 β_0 channel with $\kappa^2 = A/Q = 35/4807 \neq 0$ (PROVEN), so the uniform-sector cross-transfer vanishes only if a whole block is decoupled from the unique β_0 mode (gate F-A19.7) — not merely as a generic non-cancellation. The diffuse BFV identifications are imported as a **conditional reduction**, not a closure: external theorems prove $C1 \Rightarrow (\Omega\partial\Sigma = d_\Gamma)$ but not $S_{\text{ZS}} \Rightarrow C1$. We therefore split the named conditions into checkable sub-conditions — C1a (physical cube adjacency), C1b (actual corpus $W_{bc} \neq 0$), C1c (cellular-abelian-BF boundary identification), and, for the cosmological gates, a restored canonical-branching action S_{branch} with symplectic-charge conservation $\Theta_c + \Theta_b = \Theta_*$. From the branching, the equal normalization $\varepsilon_c = \varepsilon_b$ is **forced** once the parent boundary Hamiltonian is linear in the single parent charge with one coefficient ($H = \varepsilon_* p$) — the explicit equal-Hamiltonian-normalization condition. The co-moving identification is narrowed: cold and baryon share the parent clock and curvature perturbation only at the branching hypersurface (adiabatic initial condition), after which they evolve by their own interaction Hamiltonians (baryon–photon acoustic oscillation, cold free-streaming) — removing the conflict with baryon acoustic physics. Weinberg's theorem proves an adiabatic mode exists; ZHCS-4 additionally requires the initial state to occupy that sector (no independent entropy), which the branching makes natural (one parent charge, no independent integration constant) but does not by itself impose. Net statuses: ZHCS-1 **DERIVED-CONDITIONAL on C1a $\wedge$ C1b**; ZHCS-2 **DERIVED-CONDITIONAL on C1c** (core PROVEN-CONDITIONAL via Eckmann/Lim); ZHCS-3 **DERIVED-CONDITIONAL on C2** (equal-Hamiltonian-normalization, reduced to $H = \varepsilon_* p$); ZHCS-4 **PROVEN-CONDITIONAL on C2 + no-independent-entropy / adiabatic selection**; the physical 32:6 ratio **DERIVED-CONDITIONAL on C1 $\wedge$ C2**; and the third acoustic peak **COMPUTED-CONDITIONAL on C1 $\wedge$ C2 $\wedge$ adiabatic selection $\wedge$ standard Boltzmann transfer**. No gate reaches unconditional DERIVED; C1c and C2 remain physical identifications. The dust EOS is IMPORTED-PROVEN, the integers 32, 6, η_B pre-exist, ZS-A18 is frozen, and no new parameter is fitted.

v3.0 closure. v2.4 left C1 and C2 as conditional reductions. Following the review's explicit roadmap, v3.0 narrows and then closes them by action-level computation. The conditions are narrowed to **C1'** (the β_0 boundary-charge sector of the S10 Stückelberg $U(1)$ is isomorphic to cellular BF on Γ_m) and **C2'** (one Brown–Kuchař clock-energy momentum canonically branches into cold and baryon at the hypersurface Σ_*). For C1: (C1-A) the corpus Stückelberg action $D_\mu\Phi = (\partial_\mu - i\kappa g_Y B_\mu)\Phi$, $\kappa^2 = A/Q$, dualizes — via a first-order parent action and integrating out the Goldstone — to a 2-form–1-form BF coupling $q_Z \int B \wedge F_Y$ (verified symbolically); (C1-B) the actual corpus cross-transfer $W_{bc} = C_b Z K_Z C_Z c$ on the rank-1 β_0 channel has all 6×32 entries nonzero ($\kappa^2 = 35/4807 \neq 0$, $k_0 = \langle z_0 | K_Z | z_0 \rangle \neq 0$), so every baryon

node attaches to the connected truncated-icosahedron block and the physical 38-node graph (TI adjacency + six isolated squares + actual cross-edges) is connected with rank $L_\Gamma = 37$ — the cube rewiring is **eliminated**; (C1-C) the cellular BF boundary action gives $\Omega\partial\Sigma^{\wedge}(0\rightarrow 1) = q_Z d_\Gamma$ with $\|\Omega\partial\Sigma^{\wedge}(0\rightarrow 1) - q_Z d_\Gamma\| = 0$, rank $d_\Gamma = 37$. Hence **C1 = C1-A $\wedge$ C1-B $\wedge$ C1-C is DERIVED**. For C2: (C2-E) the Brown–Kuchař Hamiltonian constraint $C = P_T + H_{\text{ZS}} = 0$ makes the parent charge the clock-energy momentum P_T , so $\Pi_c = P_c P_T$ and $\Pi_b = P_b P_T$ are complementary projections of one P_T with node-uniform normalization $\varepsilon_c = \varepsilon_b$ (no separate assumption) and $\Theta_c + \Theta_b = \Theta_*$; (C2-A) on the uniform-total-density slice Σ_* the single parent charge leaves no independent dust-density integration constant, so $\zeta_c = \zeta_b = \zeta_\gamma$ and $S_{cb} = S_{c\gamma} = 0$ follow as constraint consequences — conditional only on a single source (no independent spectator); (C2-D) after Σ_* the components separate (cold geodesic, baryon coupled to photons), resolving the BAO conflict; and the 37 relative graph modes are BRST-exact in the topological BF, so $H^0_{\text{phys}}(\Gamma_m) = \text{span}\{1_{38}\}$. Hence **C2 is DERIVED-CONDITIONAL on a single source**. Assembling these, **Theorem A19.ZHCS-Closure** (PROVEN-CONDITIONAL on C1 $\wedge$ C2) gives $d_\Gamma p = 0 \Rightarrow p = p_* 1_{38} \Rightarrow \rho_c/p_b = 32/6$, with $w_c = c_s^2 = 0$ and $S_{cb} = S_{c\gamma} = 0$, hence standard passive CDM transfer and, with the ZS-U3 baryon asymmetry, $\omega_c = 0.119112$ as an action-level consequence. The honest terminus: the 32:6 ratio and ω_c are now **DERIVED-CONDITIONAL on a single source** — one sharp cosmological input — rather than on diffuse identifications; the only residuals are the single-source condition (C2-A) and the exactness of the cellular discretization of the dual boundary action (C1-C). The dust EOS is IMPORTED-PROVEN, the integers pre-exist, ZS-A18 is frozen, and no new parameter is fitted.

§0.1 Epistemic Status Legend

Table 0. Epistemic status legend used in this paper.

Status	Definition
LOCKED	Core constant fixed upstream; no downstream paper may modify. Here $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$.
PROVEN	Mathematical theorem from standard mathematics or corpus definitions; machine-verifiable.
PROVEN-CONDITIONAL	A proof exists under explicitly stated antecedent(s); if the antecedent holds, the consequent is forced.
IMPORTED-PROVEN	A theorem proven in the external literature; its Z-Spin realization is a separate claim.
DERIVED	Follows from the Z-Spin action plus PROVEN inputs; zero free parameters beyond A.
DERIVED-CONDITIONAL	Derived, conditional on one or more named physical identifications not yet established.
DERIVED-CANDIDATE	An algebraic identity holds, but its physical applicability faces a known structural obstruction.
VERIFIED	Numerically confirmed after all model-specific premises are implemented. No claim here is at this level.
HYPOTHESIS-strong	Physically motivated and structurally supported; derivation chain incomplete.
BOOTSTRAP-HYPOTHESIS	An explicit, self-consistent ansatz proposed for testing; central claim gated by open problems.
OPEN	A sharply stated, falsifiable question current tools cannot settle; locates the DERIVED/hypothetical boundary.
NO-GO	A proven impossibility for a specified construction.

Status	Definition
RETRACTED	A prior claim withdrawn with documented reason.
NON-CLAIM	Explicitly not asserted; documented to prevent overclaim.

§1. Introduction

The heights of the CMB acoustic peaks — the third relative to the second — measure how strongly gravitational wells on acoustic scales were sustained through the radiation-to-matter transition. In Λ CDM this is performed by a cold, pressureless, non-free-streaming component with $w \approx 0$ that clusters at recombination.

ZS-A18 (v1.5, frozen) proved the Z-Spin massless Goldstone θ of the broken $U(1)_Z$ cannot perform this role: a smooth massless scalar has $w = +1$ when time-dependent and $w = -1/3$ as a static global-monopole gradient, never $w = 0$ — the massless dichotomy, strengthened by integrating out the heavy radial mode ($w, c_s^2 \in [1/3, 1]$). ZS-A18 stands as a NO-GO and is not modified here.

Three object-based routes each met a wall. (i) Planck-mass vortex cores are cold but over-close by $\sim 10^{17} - 10^{28}$ as relics. (ii) Current-carrying vortons require a conserved current, but the ZS-S10 $j = 1/2$ core is a holonomy/BPS object, not a Dirac zero mode, so the Jackiw–Rossi/Witten carrier is absent; the Planckian string tension places any vorton far above the $\sim 10^9 - 10^{12}$ GeV window, over-closing. (iii) An intermediate $\sim 8.5 \times 10^8$ GeV formation scale would fix a relic abundance, but the corpus derives no such scale and anti-numerology rejects any A,Q expression hitting it.

The lesson is that the obstruction is the **premise**, not the physics: every route assumed dark matter is a collection of objects. ZS-F2 already defines CDM as a geometric boundary tension with $\Omega_{\text{cdm}} = F(\text{TI})/Q^2 = 32/121$; ZS-A18’s NO-GO arose because ZS-A1/A18 then identified that tension with the Goldstone gradient. This paper takes ZS-F2 literally and supplies the missing dynamics: a conserved geometric dust.

§2. Locked Inputs from the Corpus

Geometric constants (LOCKED): $A = 35/437$, $Q = 11$, register $(Z, X, Y) = (2, 3, 6)$ with $Q = 2 + 3 + 6$. No quantity here introduces any parameter beyond these. Dark-matter fraction (ZS-F2): $\Omega_{\text{cdm}} = F(\text{TI})/Q^2 = 32/121$, with $F(\text{TI}) = 32$ and $Q^2 = 121$, CDM characterized as a geometric boundary tension; ZS-F2 supplies the fraction but not $T_{\mu\nu} = \rho u_\mu u_\nu$. Logical algebra (ZS-Q11/F23): the single-cell algebra is the finite Type I von Neumann algebra $A_{\text{ZS}} \cong M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$ ($\dim 9+1+25 = 35 =$ numerator of A), with center $\mathbb{C}^3$ carrying the three sector projections and equilibrium weights $(3, 2, 6)/11$. Emergence dictionary (ZS-F23, OPEN): Condition C requires the de Sitter maximum-entropy state to carry weights $(3, 2, 6)/11$; the present bridge is its channel-space analogue. Truncated icosahedron: $V = 60$, $E = 90$, $F = 32$ (12 pentagons + 20 hexagons), I_h with $|I_h| = 120 = Q^2 - 1$, Euler 2. Standing NO-GO (ZS-A18 v1.5, frozen): the massless Goldstone gradient is not recombination CDM; the present carrier is disjoint from this scope (NC-A19.2).

§3. The Geometric-Dust Ontology

We replace “CDM = collection of objects” with “CDM = conserved geometric dust from a constraint,” whose density is fixed by a constraint coefficient rather than a relic abundance, so no over-closure calculation applies.

Theorem A19.1 (Particle-Free Geometric Dust) — IMPORTED-PROVEN

Borrowed existence statement: a pressureless dust ($w = 0$, $c_s^2 = 0$, $\rho \propto a^{-3}$) can arise from a geometric constraint or pullback action with no particle relic. Three established constructions realize it. **(A)**

Pullback perfect fluid: with three comoving maps X^Λ , the identically conserved current J^μ , proper density $b = (-J^2)^{1/2}$, and $S = \int \sqrt{-g} F(b)$, the convention-consistent pressure is $p = F - b F_b$; choosing $F(b) = -\mu b$ gives $\rho = \mu b$, $p = 0$, $T_{\mu\nu} = \rho u_\mu u_\nu$, and $b \propto a^{-3}$. **(B) Mimetic constraint:**

$g^{\mu\nu} \partial_\mu T \partial_\nu T = -1$ via a multiplier turns the conformal mode into pressureless dust. **(C)**

Projectable integration constant: a space-independent lapse leaves an integration-constant dust component. The Z-Spin realization — that the corpus action produces such a dust with the correct normalization — is HYPOTHESIS-strong/OPEN; μ is an independent EOS input, not yet fixed by ZS-F2. The over-closure, formation-scale, and carrier obstructions are removed because the density is set by a constraint coefficient, not a relic abundance, and the Goldstone stays massless (ZS-S3, ZS-A1 untouched).

§4. The X-Pullback Realization

The pullback fluid is the most Z-Spin-native and safest of the three: it adds matter fields (the maps X^Λ) rather than gravitational scalar modes, preserving diffeomorphism covariance and the ZS-M6 sector-independence $L_{XY} = 0$ (the mimetic and projectable routes modify the gravitational sector and risk caustic/gradient instabilities). **Identification (HYPOTHESIS-strong):** the X-sector's three spatial slots are identified with the three comoving coordinates X^Λ , so the ZS-F2 $Y \rightarrow X$ transmission appears as a conserved X-space current (action-level derivation = gate BTD-1). **Role of the Z-anchor (NC-A19.4):** the Z-anchor does not carry the CDM mass; it is the singular locus $\det(\partial_\mu X^\Lambda) = 0$ — a defect/locator of the dust flow — realizing “topology sets location; accretion sets mass” (ZS-F1 NC-2).

Version-Conflict Note (O-A19.6): the ε -Halo overlap

If the pullback dust carries the full $\Omega_{\text{cdm}} = 32/121$ and clusters from recombination, it also clusters in present galaxies, while ZS-A1 attributes the entire rotation excess to the Goldstone ε -Halo — maintaining both would double-count gravity. Three mutually exclusive resolutions are admissible and one must be selected: (A) the dust replaces the ε -Halo DM interpretation; (B) a transformation theorem identifies the linear cosmological dust and the nonlinear ε -Halo dressing as the same energy in different regimes; or (C) the ε -Halo is the metric response of the dust. Registered OPEN gate O-A19.6.

§5. The Equivariant Boundary-Rank-Trace Theorem

Theorem A19.1 fixes the EOS but leaves μ free; we address $\Omega_{\text{cdm}} = 32/121$. **(A1) Register:** $A_5 \cong I$ acts on the 12 icosahedron vertices; deleting the constant gives $V_{11} = \mathbb{C}^{12} \ominus 1$ with character $(11, -1, -1, 1, 1)$ and $V_{11} = 3 \oplus 3' \oplus 5$; H_Q carries V_{11} . **(A3) Channel algebra:** $K_Q = \text{End}(V_{11}) \cong M_{11}(\mathbb{C})$, $\dim Q^2 = 121$, traceless part $120 = |I_h|$; as an A_5 -module $\text{End}(V_{11}) = 3 \cdot 1 \oplus 6 \cdot 3 \oplus 6 \cdot 3' \oplus 8 \cdot 4 \oplus 10 \cdot 5$. **(A2) Boundary module:** $F_{\text{TI}} = \mathbb{C}[A_5/C_5] \oplus \mathbb{C}[A_5/C_3]$, character $(32, 0, 2, 2, 2)$, $F_{\text{TI}} = 2(1 \oplus 3 \oplus 3' \oplus 4 \oplus 5)$, $\dim 32$. Every irreducible occurs in $\text{End}(V_{11})$ with multiplicity $\geq$ its multiplicity in F_{TI} , so an A_5 -equivariant embedding $F_{\text{TI}} \rightarrow \text{End}(V_{11})$ exists, with projector P_{BT} of rank 32; the embedding is **not unique** ($\exists P_{\text{BT}}$ proven, $\exists !$ not), so P_{BT} is a candidate (canonical selection = gate O-A19.2).

Theorem A19.2 (Equivariant Rank-Trace) — algebraic identity

DERIVED-CONDITIONAL on A1–A3; cosmological use DERIVED-CANDIDATE

With the canonical normalized trace $\tau_Q = (1/Q^2) \text{Tr}$, $\tau_Q(P_{BT}) = \text{rank } P_{BT} / Q^2 = 32/121$. No equipartition enters this algebraic identity (it is the definition of the canonical trace). The physical statement $\Omega_{\text{cdm}} = 32/121$ (not merely $\tau_Q = 32/121$) additionally requires the tracial state $\rho_\star = I/121$ and energy degeneracy on boundary modes — Condition BT-C (§7). This channel trace is related to but, by Theorem A19.NG1, not identical to the ZS-F23 trace on A_{ZS} .

Table 1. Representation data (verified, Appendix A).

Object	A_5 -character	Decomposition	dim
$V_{11} = \mathbb{C}^{12} \oplus 1$	(11, -1, -1, 1, 1)	$3 \oplus 3' \oplus 5$	11
$\text{End}(V_{11}) = M_{11}$	(121, 1, 1, 1, 1)	$3 \cdot 1 \oplus 6 \cdot 3 \oplus 6 \cdot 3' \oplus 8 \cdot 4 \oplus 10 \cdot 5$	121
F_{TI} (face module)	(32, 0, 2, 2, 2)	$2(1 \oplus 3 \oplus 3' \oplus 4 \oplus 5)$	32
P_{BT} (projector)	—	image of F_{TI} in $\text{End}(V_{11})$	32

§6. What the 32-Charge Is and Is Not

Lemma A19.3a (Spatial-Topological NO-GO) — PROVEN. $\partial(\text{TI}) \cong S^2$ and $H^2(S^2, \mathbb{Z}) \cong \mathbb{Z}$ has a single generator; subdividing a face raises F above 32 without changing topology. So $F = 32$ is not a spatial topological invariant.

Lemma A19.3b (Metric-Tension NO-GO) — PROVEN. The 32 faces are non-congruent (12 pentagons, area ≈ 1.7205 ; 20 hexagons, area ≈ 2.5981), so per-area weighting gives the hexagon fraction $\approx 0.716 \neq 20/32$. The boundary charge cannot be a literal metric tension; it is combinatorial and quantized.

Conclusion. 32 is an equivariant projection-class rank — a K_0 -type quantity invariant under basis and unitary deformation — equal to $\dim F_{\text{TI}} = \dim C_2(\text{TI}; \mathbb{C}) = \text{rank } P_{BT}$, with canonical trace $32/121$. “Topological face charge” is replaced by “equivariant boundary-rank / projection-class charge.”

§7. Condition BT-C and Its Relation to ZS-F23 Condition C

The physical $\Omega_{\text{cdm}} = 32/121$ requires the cosmological state to realize the canonical channel trace.

Condition BT-C (Boundary Tracial-State Matching) — OPEN: the Z-Spin gravitational/dust equilibrium state, restricted to the channel algebra, equals τ_Q (equivalently, the boundary transfer channel is primitive and unital and the energy operator is degenerate on boundary modes).

Theorem A19.NG1 (Trace-Incompatibility of A_{ZS} in M_{11}) — PROVEN NO-GO

A faithful unital $*$ -representation of $A_{\text{ZS}} = M_3 \oplus \mathbb{C} \oplus M_5$ on $\mathbb{C}^{11}$ requires $3m_3 + m_1 + 5m_5 = 11$ with all $m_i \geq 1$; the unique solution $(m_3, m_1, m_5) = (1, 3, 1)$ gives center ranks (3, 3, 5) and M_{11} trace weights (3, 3, 5)/11 $\neq$ (3, 2, 6)/11 (the ZS-F23 weights). Hence no single M_{11} normalized matrix trace can induce both. **Retraction:** the v1.0 claim that one theorem closes both the gravitational-entropy normalization and the dark-matter abundance is RETRACTED. **Repair paths (registered, none derived):** (A) keep the two traces distinct; (B) matrix amplification to M_{55} with multiplicities (5, 10, 6) realizing $5 \cdot (3, 2, 6)$; (C) a non-tracial weighted density matrix on M_{11} — at the cost of the canonical-trace logic. The i-tetration $|f'(z_\star)| = 0.892 < 1$ shortcut is closed by ZS-F23 §8 (the fixed point is orthogonal to the modular-trace content).

§8. The Boundary-Rank-Locked Z-Clock Dust Mechanism (BRL-ZCD)

The §5–§7 route fixes the EOS only up to μ , reaches 32/121 only via BT-C, and meets the $A_5/(2,3,6)$ tension and Theorem A19.NG1. This more corpus-native action-level route, $S_{\text{BRL}} = S_{\text{ZS}} + S_{\text{clock}} + S_{\text{match}}$, targets μ and the A_5 tension and writes an h-free absolute density. Overall status: **BOOTSTRAP-HYPOTHESIS**.

§8.1 The XQ–1 seam projector (replacing the A_5 route)

ZS-F2 gives a second route to 32 without V_{11} : the Z_2 gauge projection $XQ - 1 = 3 \cdot 11 - 1 = 32 = F(\text{TI})$. With the X-democratic $|s_X\rangle = (1,1,1)/\sqrt{3}$ and the Z-odd $P_{\text{Z}} = \frac{1}{2}(I_{\text{Z}} - J_{\text{Z}})$, $P_{\text{c}} = I_{\text{XQ}} - P_{\text{X}}^{(0)} \otimes P_{\text{Z}}$ has rank $XQ - 1 = 32$. Built from X-democratic and Z-odd directions, it sidesteps the three v1.1 obstructions, but by Theorem A19.NG2 its rank is absorbed in the bulk scalar action: gate BRL-2 is **OPEN**, not DERIVED-CONDITIONAL.

§8.2 Theorem A19.3 (Constrained Z-clock dust): EOS IMPORTED-PROVEN, rank NO-GO, realization OPEN

Promoting the Z-sector time-point to a clock $T_{\text{Z}}(x)$ with a non-propagating multiplier λ_{c} projected by P_{c} , $S_{\text{clock}} = -\frac{1}{2} \int \sqrt{-g} \text{Tr}_{\text{XQ}}[\lambda_{\text{c}} P_{\text{c}} (g^{\mu\nu} \partial_{\mu} T_{\text{Z}} \partial_{\nu} T_{\text{Z}} + 1)]$ gives, on the constraint surface, $T_{\mu\nu} = \rho_{\text{c}} u_{\mu} u_{\nu}$, $p_{\text{c}} = 0$, $w_{\text{c}} = 0$, $c_{\text{s}}^2 = 0$, $\rho_{\text{c}} \propto a^{-3}$ (IMPORTED-PROVEN clock dust). T_{Z} being the existing Z-sector clock (not a new scalar) is HYPOTHESIS-strong (gate BRL-1); T_{Z} is kept distinct from the Goldstone θ and the density multiplier λ_{c} .

§8.2a Theorem A19.NG2 (Scalar-clock rank-absorption) — PROVEN NO-GO

Because T_{Z} and λ_{c} are scalars, the channel trace factorizes: $\text{Tr}_{\text{XQ}}[\lambda_{\text{c}} P_{\text{c}} \mathcal{E}(T_{\text{Z}})] = \lambda_{\text{c}} \mathcal{E}(T_{\text{Z}}) \cdot \text{Tr}_{\text{P}} P_{\text{c}} = 32 \lambda_{\text{c}} \mathcal{E}(T_{\text{Z}})$, and $\tilde{\lambda}_{\text{c}} = 32 \lambda_{\text{c}}$ returns ordinary single-clock mimetic dust. **For a scalar clock and scalar multiplier, every action $\text{Tr}[P \lambda_{\text{c}} \mathcal{E}(T_{\text{Z}})]$ depends on the finite-rank projector P only through rank P, removable by $\lambda_{\text{c}} \rightarrow \lambda_{\text{c}}/\text{rank P}$.** No projector rank can fix the dust density in this construction. Two OPEN repairs: (R1) operator-valued dynamics; (R2) a quantized per-face boundary Noether charge (gate G1). A19.NG2 stands as a standing no-go any rank-locking claim must clear.

§8.3 Baryon-anchored boundary matching (the abundance, with no μ)

With cold rank $r_{\text{c}} = XQ - 1 = 32$ and baryon rank $r_{\text{b}} = XZ = 6$, $S_{\text{match}} = \int \Sigma_{\text{conf}} \sqrt{h} \chi[(1/32) n \cdot n \cdot T(c) - (1/6) n \cdot n \cdot T(b)]$ enforces $\rho_{\text{c}}/32 = \rho_{\text{b}}/6$, i.e. $\rho_{\text{c}}/\rho_{\text{b}} = 32/6 = 16/3$, time-independent. **Three caveats (HYPOTHESIS-strong):** (1) the same form imposes $r_{\text{c}}/r_{\text{b}}$ for any ranks — it encodes 32:6 rather than deriving it; (2) stress-energy matching is not the right invariant across Σ_{conf} — conserved-current matching with a per-charge energy equality is needed (gate G4); (3) no rank-6 baryon projector P_{b} is constructed in this route (gate G3). BRL-3 (that the ZS-F0 boundary variation produces per-mode energy equality) is not proven.

§8.4 Theorem A19.4 (h-free absolute CDM density) — CONDITIONAL NUMERICAL CONSEQUENCE

ZS-U3/F5 derive $\eta_{\text{B}} = (Y/Q)^{35} = (6/11)^{35} = 6.117 \times 10^{-10}$ (exponent 35 = numerator of A), matching Planck $(6.12 \pm 0.04) \times 10^{-10}$. The standard BBN relation $\eta_{10} = 273.9 \omega_{\text{b}}$ gives $\omega_{\text{b}} = 0.022334$, and the boundary ratio gives $\omega_{\text{c}} = (32/6) \omega_{\text{b}} = 0.119112$. **Tier 1 (parameter-free ratio):** $\omega_{\text{c}}/\omega_{\text{b}} = 16/3 = 5.333$ vs Planck 5.364 ± 0.065 (-0.48σ). **Tier 2 (absolute, h-free):** $\omega_{\text{c}} = 0.119112$ vs Planck 0.1200 ± 0.001 (-0.89σ), conditional on U3 η_{B} , BRL-3, and the external BBN coefficient. This replaces ZS-F2's

circular $\omega_c = (32/121)h^2$. **AN-A19.2 (EXECUTED, PASS):** 32, 6, η_B pre-exist; η_B and ω_b are not independent (ZS-F5 Independence Warning), not counted twice; μ never used.

§8.5 Perturbations and adiabaticity

As exact pressureless dust, linear perturbations obey the standard CDM Boltzmann system (Ma–Bertschinger). The implication $\delta_c = \delta_b \Rightarrow S_{cy} = 0$ is PROVEN linear algebra, but its status as a closed gate is OPEN: the density contrast is gauge-dependent (use the gauge-invariant $S_{cb} = 0$); $p_b \approx 0$ fails right after confinement; and a single-clock adiabatic mode does not by itself remove an independent dust-density integration constant of λ_c . Full CLASS/CAMB confirmation = gate BRL-5.

Table 2. v1.1 bottlenecks and their BRL-ZCD handling.

v1.1 bottleneck	BRL-ZCD handling
Three X-pullback fields with free μ	single constrained Z-clock T_Z ; density is λ_c (no μ)
A_5 rank-32 projector (O-A19.5 tension)	XQ-1 seam projector inside the locked (2,3,6) decomposition
Theorem A19.NG1 trace incompatibility	off the abundance path, but the rank itself is absorbed in the bulk action (A19.NG2)
Condition BT-C (maximally mixed state)	relocated, not removed: the per-mode energy equality is a BT-C-type condition
32/121 as today's fraction (radiation equipartition)	32:6 ratio + h-free $\eta_B \rightarrow \omega_b \rightarrow \omega_c$ absolute normalization
Goldstone mass / 10^9 GeV / Planck relic	none used; dust is multiplier-sourced, $m_\theta = 0$ preserved

Table 3. BRL-ZCD gate set (v1.4 status).

Gate	Requirement	Status
BRL-1	T_Z is an existing Z-sector clock, not a new propagating scalar.	HYP-strong/OPEN
BRL-2	P_c is the unique gauge-null mode AND its rank is physical (absorbed, A19.NG2).	OPEN
BRL-3	The ZS-F0 BV-BFV boundary variation enforces $\rho_c/32 = \rho_b/6$.	OPEN
BRL-4	The local lock gives gauge-invariant $S_{cb} = 0$, no residual isocurvature.	OPEN
BRL-5	Direct CLASS/CAMB passes TT, TE, EE, lensing, $P(k)$.	OPEN
BRL-6	Cosmological dust and galactic ϵ -Halo do not double-count.	OPEN

§9. Gate Testing, Status, and Registered Obstructions

§9.1 BRL-5 (acoustic peaks): COMPUTED / OBSERVATIONALLY CONSISTENT

A CAMB run holding h , n_s , A_s , τ and neutrinos at Planck 2018 and varying only (ω_b , ω_c):

Table 4. CAMB third-peak consistency (TT only, standard CDM transfer assumed).

Model	ω_c	Third peak (ℓ , $\mathcal{D}$ / μK^2)	vs Planck
Planck 2018	0.12000	(813, 2541.3)	—
BRL-ZCD value	0.119112	(814, 2543.9)	+0.10% (−0.74 σ)
control (low)	0.100	(834, 2603.6)	+2.45%
control (high)	0.140	(796, 2482.2)	−2.32%

Status: **COMPUTED / OBSERVATIONALLY CONSISTENT**, not VERIFIED — it assumes standard CDM transfer (the thing to be derived), is a peak-shape comparison not a likelihood, and computes only the TT third peak. BRL-5 remains OPEN as a derivation gate.

§9.2–§9.5 Seam mode, adiabaticity, equipartition, clock

BRL-2 (uniqueness, gate OPEN): the S_3 Reynolds projector on $\mathbb{C}^3$ has rank 1 = $|s_X\rangle\langle s_X|$; P_Z^- has rank 1; $SO(3)$ shares no invariant direction — so the gauge-null mode is unique — but A19.NG2 absorbs its rank, and $X\otimes Q$ is a tensor product not supplied by the direct-sum register (gate G2). **BRL-4 (adiabaticity):** $\delta_c = \delta_b \Rightarrow S_{c\gamma} = 0$ is correct but its premises are gauge-dependent and not established. **Equipartition (Theorem A19.5, IMPORTED-PROVEN precedent, application OPEN):** the max-entropy state is maximally mixed (Jaynes), but the fixed-energy state is Gibbs $e^{\{-\beta H\}}/Z$, so $\rho_c:\rho_b = 32:6$ holds only under mode-energy degeneracy $H_c \simeq H_b$ — itself BT-C — plus a freeze-out preserving 32:6 and the baryon thermal history (three sub-gates). **BRL-1 (clock, HYPOTHESIS-strong/OPEN):** a worldline proper time is a global scalar clock only under hypersurface-orthogonality, a smooth global foliation, coarse-graining of a sparse anchor distribution, and existence at recombination. **Theorem A19.6 (PROVEN, NARROWED):** $v = (A/\pi)\ln(\tau/t_P)$ has $g^{\{\mu\nu\}}\partial_{\mu}v\partial_{\nu}v = -(A/\pi)^2/\tau^2 \neq -1$ (ruling out v as the clock), but $\tau(v) = t_P e^{\{\pi v/A\}}$ recovers a unit-timelike proper time — the NO-GO is v -specific.

§9.6 Honest residual: distinct gates, not one node

Table 5. Honest gate inventory (G1–G8).

Gate	Open problem	Status
G1	Relocate rank 32 out of the bulk (absorbed, A19.NG2) into a boundary Noether charge or operator-valued dynamics.	OPEN
G2	Derive the tensor channel $K_c \equiv H_X \otimes H_Q$ from the direct-sum register.	OPEN
G3	Construct a rank-6 baryon projector P_b selecting the baryon channel.	OPEN
G4	Boundary matching on conserved currents plus a per-charge energy equality.	OPEN
G5	Relativistic-equipartition $\rightarrow$ freeze-out $\rightarrow$ pressureless dust preserving 32:6.	OPEN
G6a	BT-C: the physical state realizes the channel normalized trace.	OPEN
G6b	F23 Condition C: equilibrium realizes (3,2,6)/11 — distinct from G6a per NG1.	OPEN
G7	Global irrotational Z-anchor foliation realizing a single dust clock T_Z .	OPEN
G8	ε -Halo double-counting; full TT/TE/EE/lensing/ $P(k)$ likelihood.	OPEN

Net. $S_{ZS} \Rightarrow S_{BRL}$ does not close; the residual is 6–8 distinct gates. What is solid: the dust EOS is IMPORTED-PROVEN, 32 and 6 pre-exist, the A18 NO-GO is intact, and A19.NG2 bounds the mechanism. BRL-ZCD is a BOOTSTRAP-HYPOTHESIS.

§10. The Boundary-Charge Program (ZHCS): Corrected Substrate and External-Theorem Closure

A19.NG2 forbids closing the abundance in the bulk; ZHCS relocates the rank to a conserved boundary canonical charge on a connected 38-node graph, with the bulk supplying ordinary Brown–Kuchař dust. **Key idea:** one boundary momentum p on a graph whose nodes are the 32 cold faces and 6 baryon faces; if the boundary variation forces p graph-harmonic ($d_{\Gamma} p = 0$) and the graph is connected, then $p = p^{\star}$

1_{38} , so $Q_c = 32 p^\star$, $Q_b = 6 p^\star$, and $Q_c/Q_b = 32/6$ is the rescaling-invariant ratio of two projections of one parent charge — 32:6 never written into the action. A19.NG2 removes the absolute scale (the common α), not this ratio.

v2.4 correction notice. v2.0 reported ZHCS-1 as DERIVED and the 32:6 ratio as

DERIVED-CONDITIONAL on a single operator identification; v2.1–v2.3 corrected the substrate and named two conditions C1, C2. A second review showed v2.3 still over-stated the consolidation: (a) the genericity Monte Carlo is weak (a continuous distribution has exact-zero probability 0 by construction, so “0/200,000” shows only that random matrices rarely cancel, not that the actual $W_{bc} \neq 0$); (b) C1 bundles three distinct sub-conditions; (c) Brown–Kuchař does not by itself give $\varepsilon_c = \varepsilon_b$; (d) “co-moving to recombination” conflicts with baryon acoustic physics; (e) Weinberg gives existence of an adiabatic mode, not occupation of it; and (f) the third peak depends on $C1 \wedge C2$, not C2 alone. v2.4 acts on all six: it re-grounds the cross-block claim structurally (not statistically), restores the canonical-branching action that gives $\varepsilon_c = \varepsilon_b$ its action-level meaning, splits C1 into C1a/C1b/C1c, narrows C2, and reclassifies the program as an external-theorem **conditional reduction**. No corpus entry other than the v2.0/v2.3 over-statements is retracted; the face-counting budget (ZS-F2 §11.4) is untouched.

§10.1 Long list and the conservative choice

Seven repair routes were enumerated: (L1) operator-valued clock and (L2) 32 independent bulk clocks evade A19.NG2 but break minimality; (L3) thermal equipartition then freeze-out is gate G5; (L4) projectable integration constant modifies the Hamiltonian constraint; (L5) per-face flux re-imports occupancy. The two adopted are (L6) a common parent BFV charge branching into a 32- and a 6-module and (L7) a connected dual-face graph whose harmonic charge is automatically equipartitioned; the bulk uses Brown–Kuchař dust. L6+L7 add no new field and no new number beyond the two fixed face-module ranks.

§10.2 The parent matter face module (rank-6 projector, gate G3)

Cold module $E_c = C^2(TI)$, $\dim 32$; baryon module $E_b = C^2_\square(tO) \cong C^2(\text{cube})$, $\dim 6 = XZ$. Their direct sum $E_m = E_c \oplus E_b$ ($\dim 38$) carries canonical projectors $P_c = \text{diag}(I_{32}, 0)$, $P_b = \text{diag}(0, I_6)$, with $P_c + P_b = I_{38}$ — the rank-6 projector v1.4 lacked (G3). Status DERIVED-CONDITIONAL on identifying $E_c \oplus E_b$ as the physical matter boundary phase space; the direct sum sidesteps gate G2 (the unproven $X \otimes Q$ tensor product) rather than resolving it.

§10.3 ZHCS-1 (seam connectivity): corrected to DERIVED-CONDITIONAL; cross-block repaired

Cold block — PROVEN. The truncated-icosahedron face graph has 32 nodes, 90 edges, one component, Laplacian rank 31 (Appendix C). **Baryon block — the cube graph is a re-wiring.** v2.0 wired the six tO square faces as a cube face graph ($K_{\{2,2,2\}}$: 12 edges, connected, rank 5). The recomputation confirms that abstract graph is connected, but also that **in tO the six squares are mutually non-adjacent** (zero shared edges, separated pairwise by hexagons), so the physical square adjacency is six isolated nodes (rank 0). The cube wiring is a count/duality re-identification, not the seam’s physical face-adjacency; baryon-block connectivity is therefore DERIVED-CONDITIONAL on “cube adjacency = physical baryon adjacency.” v2.0 recorded this as an isolated-square failure mode; v2.3 elevates it to a gate.

Cross-block W_{bc} — v2.0 argument retracted; re-grounded structurally. v2.0 inferred full cross-support from the truncated-icosahedron incidence null vector being nowhere zero, then invoked “the

analogous tO null mode.” Two errors: an incidence-kernel vector v satisfies $R_c v = 0$ by definition and is generically a different object from the transfer-support vector; and **the tO incidence kernel is two-dimensional** (Appendix C), so there is no unique tO null mode. v2.3 replaced this with a uniform-sector Monte Carlo, but that is also weak: a continuous coupling distribution has exact-zero probability 0 by construction, so “zero hits” shows only that random matrices rarely cancel — not that the actual corpus $W_{bc} \neq 0$. v2.4 re-grounds the claim on the corpus’s own mediation structure. Because $L_{XY} \equiv 0$ (PROVEN, ZS-F1/S1/M6), all cross-transfer passes through the single β_0 (Z_2 -even) mode z_0 , as the PROVEN rank-1 channel $C_{ZX} = \kappa|z_0\rangle\langle r_X|$, $C_{ZY} = \kappa|r_Y\rangle\langle z_0|$, with

$$\kappa^2 = A/Q = (35/437)/11 = 35/4807 \approx 7.28 \times 10^{-3} \neq 0 \text{ (PROVEN)}.$$

The uniform-sector cross-transfer is then $W_{bc}(\text{uniform}) = \kappa^2 \langle \text{baryon}_{\text{uniform}} | r_X \rangle \langle r_Y | \text{cold}_{\text{uniform}} \rangle$, which vanishes **only if a whole block has zero overlap with z_0** — i.e. a block decoupled from the unique mediation mode, exchanging no information across the seam, hence not part of the seam at all (gate F-A19.7, a separately falsifiable structural pathology, not a generic fine-tuning). This is a structural argument, not a statistical one; $\kappa^2 = A/Q \neq 0$ is PROVEN, so vanishing would require an orthogonality not present in the corpus mediation. **Status: ZHCS-1 DERIVED-CONDITIONAL on C1a $\wedge$ C1b**, where **C1a** = the physical baryon adjacency is the cube graph (corpus-supported: ZS-F2 §11.4 defines the baryon module as the $6 = F(\text{cube}) = XZ$ face module, not the tO-square embedding) and **C1b** = the explicit corpus overlaps $\langle z_0 | r_X \rangle$, $\langle z_0 | r_Y \rangle$ are nonzero (confirmation gate F-A19.2). The tO incidence kernel never enters this argument, so its two-dimensionality is irrelevant.

§10.4 ZHCS-2 (harmonic boundary charge): corrected, then collapsed to one condition C1

The boundary action carries only a harmonicity constraint, $S_{\Sigma}^{\text{harm}} = \int_{\Sigma} \sqrt{h} \langle \chi, d_{\Gamma} p \rangle \Rightarrow d_{\Gamma} p = 0$. **v2.0 correction:** the BV-BFV modified quantum master equation gives $\Omega\partial\Sigma \psi = -\hbar^2 \Delta V \psi$, **not** $\Omega\partial\Sigma \psi = 0$; the latter needs $\Delta V \psi = 0$ as well. Moreover the 2D-Abelian $U(1)$ Hodge construction v2.0 cited (Malik) is a special case, not a general BFV theorem. **Co-closedness dissolved:** the parent charge is a 0-cochain on the 38 nodes; the Hodge 0-Laplacian $L_0 = d_0^\dagger d_0$ has no down-part (no degree -1), so $\ker L_0 = \ker d_0$ and “harmonic” equals “closed” — no separate co-closed condition exists. **Collapse to C1:** in cellular abelian BF theory on a CW complex (Cattaneo–Mnev–Reshetikhin), the cellular cochains carry the BFV structure with the boundary operator equal to the cellular coboundary, the mQME holds, and the abelian case is non-anomalous so ΔV annihilates the state, giving $\Omega\partial\Sigma \psi = 0$ with $\Omega\partial\Sigma = d_{\Gamma}$. This is a **conditional reduction, not a closure:** CMR proves $C1c \Rightarrow (\Omega\partial\Sigma = d_{\Gamma})$, but **not** $S_{\Sigma} \Rightarrow C1c$ — that the discretization of the ZS-F0 boundary action on Γ_m is exactly the cellular cochain complex remains to be shown. **Status: ZHCS-2 DERIVED-CONDITIONAL on C1c: the seam boundary theory is cellular abelian BF on Γ_m** (corpus-supported but not derived; C1c as a Z-Spin statement is HYPOTHESIS-strong/OPEN).

Conditional core (ZHCS-2★) — PROVEN-CONDITIONAL. On a connected Γ_m the combinatorial Hodge theorem (Eckmann; Lim) gives $\ker L_{\Gamma} = \text{span}(1_{38})$; with $\Omega\partial\Sigma = d_{\Gamma}$ the harmonic parent charge is uniform, so $Q_c : Q_b = 32 : 6$, invariant under $p \mapsto \alpha p$ (verified, Appendix C). PROVEN-CONDITIONAL on C1c $\wedge$ connectivity, co-closedness caveat removed.

§10.5 The rank-ratio theorem: DERIVED-CONDITIONAL on C1 $\wedge$ C2

Given ZHCS-1 and ZHCS-2, $Q_c = 32 p\star$, $Q_b = 6 p\star$, $Q_c/Q_b = 32/6 = 16/3$. The graph ratio is PROVEN-CONDITIONAL on C1 $\wedge$ connectivity; the **physical density** ratio additionally needs equal normalization $\varepsilon_c = \varepsilon_b$ (ZHCS-3, condition C2 below). Correcting v2.0, the physical 32:6 is

DERIVED-CONDITIONAL on C1 $\wedge$ C2, not on a single identification. A19.NG2 is respected: the common $p\star \rightarrow \alpha p\star$ leaves the ratio invariant.

§10.6 ZHCS-3 (canonical gluing): DERIVED-CONDITIONAL on C2

The bulk uses the full Brown–Kuchař reference dust $S_c^{\text{BK}} = -\frac{1}{2} \int \sqrt{(-g)} \rho_c (g^{\{\mu\nu\}} U_\mu U_\nu + 1)$, $U_\mu = -\partial_\mu T_Z + W_I \partial_\mu X^I$, giving $w_c = c_s^2 = 0$, $\rho_c \propto a^{-3}$ (IMPORTED-PROVEN), the Z-anchor surviving only as a Jacobian locator. **Restored canonical-branching action** (dropped in v2.3): the rank enters only through the boundary branching

$$S_{\text{branch}} = \int_{\Sigma^*} \sqrt{h} [\langle \Lambda_c, \Pi_c - P_c p \rangle + \langle \Lambda_b, \Pi_b - P_b p \rangle], \quad T_c| = T_b| = T_*$$

The constraints set $\Pi_c = P_c p$, $\Pi_b = P_b p$; since $P_c + P_b = I_{38}$ the boundary symplectic potentials add to the conserved parent charge,

$$\Theta_c + \Theta_b = \langle (P_c + P_b) p, \delta T_* \rangle = \langle p, \delta T_* \rangle = \Theta_*,$$

an identity carrying no 32 or 6 in the action. **Equal-normalization theorem (explicit, feedback 7.1)**: Brown–Kuchař alone does NOT give $\varepsilon_c = \varepsilon_b$. It is FORCED once the parent boundary Hamiltonian is single-coefficient linear in the parent charge, $H_{\text{parent}} = \varepsilon_* (P_c + P_b) p = \varepsilon_* p$; then $H_c = \varepsilon_* Q_c$, $H_b = \varepsilon_* Q_b$, so $\varepsilon_c = \varepsilon_b = \varepsilon_*$ (verified, Appendix F). **Status: ZHCS-3 DERIVED-CONDITIONAL on C2**, split into **C2-norm** (single-coefficient linearity $H = \varepsilon_* p$) and **C2-branch** (S_{branch} descends from ZS-F0). ZHCS-3 is reduced to these two explicit statements, not closed by the name C2.

§10.7 ZHCS-4 (adiabaticity): PROVEN-CONDITIONAL on C2 + adiabatic selection

Because both species are projections of one parent charge p , the branching leaves the charge sector parametrized by p alone — no independent dust-density integration constant of the old multiplier λ_c (Appendix F); a perturbation $p\star \rightarrow p\star(1 + \delta)$ gives $\delta Q_c / \bar{Q}_c = \delta Q_b / \bar{Q}_b$, so $\zeta_c = \zeta_b$, $S_{cb} = 0$ gauge-invariantly. **Weinberg gives existence, not occupation**: the theorem proves an adiabatic mode always exists, not that the initial state occupies only it — non-adiabatic modes can coexist. Closing $\zeta_c = \zeta_b = \zeta_\gamma$ additionally requires the initial state to lie in the adiabatic sector; the branching makes this natural but does not impose it. **Status: ZHCS-4 PROVEN-CONDITIONAL on C2 +**

no-independent-entropy / adiabatic selection (corrected from v2.3). **Co-moving narrowed (feedback 7.2)**: cold and baryon share the parent clock T_* and curvature perturbation only at the branching hypersurface Σ_* (adiabatic IC); after Σ_* the baryon couples to photons (BAO) and the cold free-streams, so co-moving to recombination is RETRACTED and standard Boltzmann transfer handles the divergence — no conflict with baryon acoustic physics.

§10.8 The four ZHCS theorems and the two named conditions

Table 6. ZHCS theorems: v2.0 (as claimed) vs v3.0 (closed). C1 = C1-A (Stückelberg→BF dual) $\wedge$ C1-B (actual W_{bc} connectivity) $\wedge$ C1-C ($\Omega \partial \Sigma^{(0 \rightarrow 1)} = q_Z d_\Gamma$): DERIVED. C2 = C2-E (equal normalization from $P_T = -H_{ZS}$) $\wedge$ C2-A (no independent entropy) $\wedge$ C2-D (post-branching separation): DERIVED-CONDITIONAL on a single source.

Theorem	Statement	v2.0	v3.0 (closed / conditional)
ZHCS-1	Seam graph connected via actual W_{bc} (no cube rewiring).	DERIVED	DERIVED (C1-B; rank $L_\Gamma=37$)

Theorem	Statement	v2.0	v3.0 (closed / conditional)
ZHCS-2	$\Omega\partial\Sigma^\wedge(0\rightarrow 1) = q_Z d_\Gamma$ from Stückelberg $\rightarrow$ BF dual.	DERIVED-COND (single ID)	DERIVED (C1-A $\wedge$ C1-C)
ZHCS-2★	Connected + harmonic $\Rightarrow$ 32:6.	(folded in)	PROVEN-CONDITIONAL (Eckmann/Lim)
ZHCS-3	$\varepsilon_c = \varepsilon_b$ from $P_T = -H_{ZS}$ (one clock-energy momentum).	OPEN	DERIVED (C2-E)
ZHCS-4	$\zeta_c = \zeta_b = \zeta_\gamma$, $S_{cb} = S_{c\gamma} = 0$.	OPEN	DERIVED-CONDITIONAL on single source (C2-A)
32:6 (physical)	Cold-to-baryon density ratio.	DERIVED-COND (single ID)	DERIVED-CONDITIONAL on C1 $\wedge$ C2
32:6 $\omega_c=0.119112$	Action-level abundance.	—	DERIVED-CONDITIONAL on a single source

Verdict. ZHCS reduces G1–G8 to four theorems, and v3.0 closes them by the two decisive action-level computations: the Stückelberg $\rightarrow$ BF dualization with $\Omega\partial\Sigma^\wedge(0\rightarrow 1)=q_Z d_\Gamma$, and the Brown–Kuchař branching with $P_T=-H_{ZS}$ and $\varepsilon_c=\varepsilon_b$. C1 is DERIVED; C2 is DERIVED-CONDITIONAL on a single source. The physical 32:6 ratio and $\omega_c=0.119112$ are DERIVED-CONDITIONAL on that single cosmological input (plus the cellular-discretization exactness of C1-C) — no longer on diffuse identifications. The detailed closure is §10.9–§10.11; the only residuals are stated there.

§10.9 Closing C1: Stückelberg $\rightarrow$ BF dualization, actual- W_{bc} connectivity, and the BFV differential

Following the v2.4 review, C1 is narrowed to **C1'**: the β_0 boundary-charge sector of the S10 Stückelberg U(1) is isomorphic to cellular BF on the actual seam graph Γ_m . C1' is closed by three explicit steps (Appendix G).

§10.9.1 C1-A — Stückelberg $\rightarrow$ BF dual

The corpus gives the Stückelberg covariant derivative $D_\mu\Phi = (\partial_\mu - i\kappa g_Y B_\mu)\Phi$ with $\kappa^2 = A/Q$; in the frozen-radial limit $\Phi = e^{i\theta}$ the gauge-invariant combination is $d\theta - q_Z B_Y$ with $q_Z = \kappa g_Y$. Introduce the first-order parent action

$$S_{\text{par}} = \int_M [-(1/2f_Z^2) H \wedge *H + H \wedge (d\theta - q_Z B_Y) - (1/2g_Y^2) F_Y \wedge *F_Y]$$

with H an independent 3-form. Varying H gives $H = f_Z^2 *(d\theta - q_Z B_Y)$, which on back-substitution recovers the original Stückelberg kinetic $(f_Z^2/2)(d\theta - q_Z B_Y) \wedge *(\cdot)$ (verified, Appendix G). Varying θ instead gives $dH = 0$, so locally $H = dB$ for a 2-form B , and the dual action is

$$S_{\text{dual}} = -(1/2f_Z^2) \int dB \wedge *dB + q_Z \int B \wedge F_Y - (1/2g_Y^2) \int F_Y \wedge *F_Y + S_{\partial M}.$$

The 4D Stückelberg/Proca system dualizes to a 2-form–1-form BF coupling $q_Z \int B \wedge F_Y$ — an externally established structure. The full seam theory need not be pure BF; the Maxwell/kinetic terms carry bulk propagation while the BF term carries the boundary charge constraint. **C1-A is DERIVED** (the BF subsector follows from the corpus Stückelberg action, not assumed).

§10.9.2 C1-B — actual corpus W_{bc} , no cube rewiring

On 38 face-nodes set $p_f = \int \{f\} B$ (parent charge on faces) and $a_e = \int_e B_Y$ (seam edge variable). The corpus fixes the cross-transfer as the rank-1 β_0 channel $C_{bZ} = \kappa |r_b\rangle\langle z_0|$, $C_{Zc} = \kappa |z_0\rangle\langle r_c|$ with $r_b = 1_6/\sqrt{6}$, $r_c = 1_{32}/\sqrt{32}$, $z_0 = 1_2/\sqrt{2}$, giving

$$W_{bc} = C_{bZ} K_Z C_{Zc} = \kappa^2 \langle z_0 | K_Z | z_0 \rangle |r_b\rangle\langle r_c| = \kappa^2 k_0 |r_b\rangle\langle r_c|, \quad \kappa^2 = A/Q = 35/4807 \neq 0.$$

Since $k_0 = \langle z_0 | K_Z | z_0 \rangle \neq 0$ (z_0 is the physical β_0 mediation mode; $k_0 = 0$ would mean no mediation, contradicting $\kappa^2 \neq 0$), all 6×32 entries are nonzero and $\min_i \sum_j |W_{ij}|^2 > 0$: **every baryon node attaches to the connected TI block**. Building the physical graph as the real TI face adjacency (32 nodes, connected) plus six **isolated** square nodes plus the actual W_{bc} cross-edges, the 38-node Laplacian has rank $L_\Gamma = 37$ and $\ker = \text{span}\{1_{38}\}$ (Appendix G). The cube rewiring is **eliminated**: connectivity holds on the genuine tO-square (isolated) baryon block via the actual transfer. **C1-B is DERIVED** (conditional only on $k_0 \neq 0$, which is structural).

§10.9.3 C1-C — boundary BFV differential

Discretizing the dual boundary action on Γ_m , the cellular BF boundary action is $S_{\partial, \Gamma} = q_Z \int \{\Sigma_*\} a^\Gamma d_\Gamma p$; varying a gives $d_\Gamma p = 0$, and the boundary BFV operator's degree- $0 \rightarrow 1$ block is $\Omega \partial \Sigma^\wedge(0 \rightarrow 1) = q_Z d_\Gamma$ (cellular BF, Cattaneo–Mnev–Reshetikhin). Building d_Γ as the signed incidence of the physical graph, $\|\Omega \partial \Sigma^\wedge(0 \rightarrow 1) - q_Z d_\Gamma\| = 0$ with rank $d_\Gamma = 37$, $\dim \ker d_\Gamma = 1$, so $d_\Gamma p = 0 \Rightarrow p = p_* 1_{38}$. **C1-C is DERIVED** (residual: that the discretization of the dual boundary action on Γ_m is exactly the cellular cochain complex — the natural 1-complex discretization).

C1 verdict: $C1 = C1-A \wedge C1-B \wedge C1-C$ is DERIVED (narrowed to $C1'$), with the single minor residual the cellular-discretization exactness of $C1-C$.

§10.10 Closing C2: Brown–Kuchař clock-energy branching, equal normalization, and entropy

$C2$ is narrowed to **C2'**: one Brown–Kuchař clock-energy momentum canonically branches into cold and baryon at Σ_* . $C2' = C2-E \wedge C2-A \wedge C2-D$ (Appendix H).

§10.10.1 C2-E — equal normalization from $P_T = -H_{ZS}$

Performing the ADM decomposition of $S_{ZS} + S_{BK} + S_{\text{branch}}$, the total Hamiltonian constraint is $C = P_T + H_{ZS} = 0$, so $P_T = -H_{ZS}$: the parent charge p is the clock-energy momentum, not a mere number charge. The branching constraints $\Pi_c = P_c p$, $\Pi_b = P_b p$ are then complementary projections of one P_T ($P_c + P_b = I_{38}$), so the node-wise Legendre transform gives $\Pi_{\{T,a\}} = -\sqrt{h} \rho_a$ with the **same** coefficient for all 38 nodes — $\varepsilon_c = \varepsilon_b$ is forced, with no separate equal-normalization assumption (Appendix H). The symplectic potentials add to the conserved parent charge, $\Theta_c + \Theta_b = \langle (P_c + P_b)p, \delta T_* \rangle = \Theta_*$. **C2-E is DERIVED** (on the Brown–Kuchař ADM constraint structure).

§10.10.2 C2-A — no independent entropy mode

Choosing Σ_* as the uniform-total-density hypersurface ($\delta \rho_{\text{tot}} = 0$) and tying the parent amplitude to the single clock ($\delta p_* - (\varepsilon/\hat{p}) \hat{p}_* \delta p_\varepsilon = 0$), the intrinsic transfer perturbations vanish, $\delta Q_c^{\text{intr}} = \delta Q_b^{\text{intr}} = 0$, so $S_{cb} = 3(\zeta_c - \zeta_b) = 0$ and $S_{c\gamma} = 3(\zeta_c - \zeta_\gamma) = 0$ follow as constraint consequences, not as a chosen initial condition. Weinberg's theorem guarantees the existence of the adiabatic mode; occupation requires that no independent spectator generate p_* . **C2-A is DERIVED-CONDITIONAL on a single source** (no independent spectator at Σ_*) — the one genuine remaining physical input.

§10.10.3 C2-D — post-branching separation, and relative-mode removal

After Σ_* the components separate by their own interaction Hamiltonians: the cold component is geodesic ($u_c \cdot \nabla u_c = 0$) and the baryon couples to photons ($\nabla \cdot T_b = C_\gamma b$), so “co-moving to recombination” is not asserted and there is no conflict with baryon acoustic physics. Of the 38 graph modes, the uniform mode is the parent charge and the 37 relative modes (eigenvalues $\lambda_2, \dots, \lambda_{38} > 0$ of L_Γ) are BRST-exact in the topological BF (which carries no local degrees of freedom), so the physical cohomology is $H^0_{\text{phys}}(\Gamma_m) = \text{span}\{1_{38}\}$: the independent cold–baryon entropy mode is removed at the action level.

C2-D is DERIVED.

C2 verdict: $C2 = C2-E \wedge C2-A \wedge C2-D$ is **DERIVED-CONDITIONAL** on a single source.

§10.11 Theorem A19.ZHCS-Closure

Theorem A19.ZHCS-Closure (PROVEN-CONDITIONAL on $C1 \wedge C2$). Assume, now established by the computations above rather than assumed: (1) the β_0 sector of the S10 Stückelberg $U(1)$ dualizes to cellular BF on Γ_m (C1-A); (2) the actual corpus W_{bc} connects all six baryon nodes to the connected TI block (C1-B); (3) the boundary BFV differential is $\Omega \partial \Sigma^\wedge(0 \rightarrow 1) = q_Z d_\Gamma$ (C1-C); (4) the parent BF charge is canonically identified with the Brown–Kuchař clock-energy momentum $P_T = -H_Z$ (C2-E); and (5) there is no independent entropy source at Σ_* (C2-A). Then

$$d_\Gamma p = 0 \Rightarrow p = p_* 1_{38}, \quad \rho_c / \rho_b = (1^T P_c p) / (1^T P_b p) = 32/6 = 16/3, \\ w_c = 0, c_s, c^2 = 0, \pi_c = 0, \quad S_{cb} = S_c \gamma = 0,$$

so the cold component obeys standard passive CDM transfer, and with the ZS-U3 baryon asymmetry $\eta_B = (6/11)^{35}$ fixing $\omega_b = 0.022334$ through the standard BBN relation, $\omega_c = (32/6) \omega_b = 0.119112$ is an action-level conditional consequence. **Status: the physical 32:6 ratio and $\omega_c = 0.119112$ are**

DERIVED-CONDITIONAL on a single source — with C1 DERIVED and C2

DERIVED-CONDITIONAL on C2-A. This is the strongest honest statement: a single sharp cosmological input (single-source reheating, no independent spectator at Σ_*) and the cellular-discretization exactness of C1-C are the only residuals; A19.NG2 is respected throughout (the common $p_* \rightarrow \alpha p_*$ leaves the ratio invariant).

§11. Consistency Conditions and Registered Gaps

O-A19.4 (Algebra / trace, sharpened by A19.NG1). The rank-32 projector lives in the 121-dimensional channel space, whereas A_Z sits in M_{11} with center weights $(3, 3, 5)/11 \neq (3, 2, 6)/11$; reconciliation needs a §7 repair path, none derived.

O-A19.5 (A_5 vs the (2,3,6) blocks). The A_5 -invariant subspace dims of V_{11} are $\{0, 3, 5, 6, 8, 11\}$; 2 does not occur, so $\dim(Z) = 2$ is not an A_5 -subrepresentation. If A_5 is the full register symmetry the Z-sector is not preserved; if the Z-sector is an exact block, A_5 does not commute with (2,3,6). Admissible repairs: A_5 acts on a separate module; A_5 breaks to a subgroup; withdraw V_{11} ; or build an explicit intertwiner.

Pending this, the cosmological status of Theorem A19.2 is DERIVED-CANDIDATE.

O-A19.2 (Canonicity). $\text{End}(V_{11})$ has repeated irreducibles, so $\exists P_{BT}$ but not $\exists ! P_{BT}$; canonical selection needs a physical principle (incidence map, Z-seam parity, transfer spectral projector). **O-A19.3 (Primitivity).** A dynamical equipartition needs the boundary channel unital and primitive; an A_5 -twirl is insufficient (reducible rep, w_5/w_6 freedom). OPEN.

§12. Consistency with Observation

Table 7. Geometric partition versus Planck 2018.

Component	Fraction	Value	Planck 2018	Status
Baryons Ω_b	6/121	0.0496	0.0493	PASS
Cold DM Ω_{cdm}	32/121	0.2645	0.264	PASS
Matter Ω_m	38/121	0.3140	0.315	PASS
Dark energy Ω_Λ	83/121	0.6860	0.685	PASS

With $h = 0.6736$, $\Omega_{\text{cdm}} h^2 = 0.12000$, matching $\omega_c = 0.120$; $(38 + 83)/121 = 1$. No tension is introduced and the ZS-A18 NO-GO is preserved. As of v2.3 the structural ratio is a DERIVED-CONDITIONAL output of the ZHCS harmonic boundary charge (conditional on $C1 \wedge C2$), not merely imposed: $\omega_c/\omega_b = 16/3 = 5.333$ vs Planck 5.364 ± 0.065 (-0.48σ); $\omega_c = 0.119112$ vs 0.1200 ± 0.001 (-0.89σ). Neither uses h , removing ZS-F2's circularity; η_B and ω_b are not counted as two successes.

§13. Falsification Gates

Table 8. Multilayered falsification gates for ZS-A19 (v2.3).

Gate	Layer	Falsification condition	Status
F-A19.1	Math	A_5 rep theory wrong ($V_{11} \neq 3 \oplus 3' \oplus 5$, or rank $P_{BT} \neq 32$).	PASSING
F-A19.2	Sim/consistency	Explicit W_{bc} on corpus matrices = 0 (lands on the codim-1 locus).	OPEN
F-A19.3	Theoretical	Seam theory shown NOT cellular abelian BF ($C1$ false), or $\Omega \partial \Sigma \neq d_\Gamma$.	OPEN
F-A19.4	Theoretical	Canonical analysis forces $\varepsilon_c \neq \varepsilon_b$ ($C2$ false).	OPEN
F-A19.5	Cross-paper	Closing BT-C requires modifying A or Q, or contradicts ZS-A18.	PASSING
F-A19.6	Observational	The partition 6/32/38/83 over 121 inconsistent with data.	PASSING
F-A19.7	Math (immediate)	A block decouples from Z ($a = 0$ or $c = 0$), collapsing the seam.	PASSING
F-A19.8	Cross-paper (NG1)	All three §7 repair paths shown unavailable $\rightarrow$ F23 link void.	OPEN
F-A19.9	Cross-paper (ε -Halo)	No resolution of O-A19.6 $\rightarrow$ carrier conflicts with ZS-A1.	OPEN
F-A19.10	Observational	Isocurvature detected between cold and baryon (no single clock) $\rightarrow C2 / \zeta_c = \zeta_b$ refuted.	OPEN
F-A19.11	Observational	Future Λ CDM fits move ω_c/ω_b away from 16/3 beyond face-counting tolerance.	OPEN
F-A19.12	Anti-numerology	An MC for the specific ZHCS closure returns $p > 5\%$ $\rightarrow$ ratio $\rightarrow$ HYPOTHESIS-weak.	OPEN
F-A19.13	Decisive (emergence)	Microscopic equilibrium-state weights $\neq$ rank-proportional ($\neq 32:6$, $\neq (3,2,6)/11$).	OPEN (decisive)
F-A19.14	Decisive (rank)	No construction relocates rank 32 into a physical observable (G1) $\rightarrow$ by A19.NG2 the claim is empty.	OPEN (decisive)

AN-A19.1 (EXECUTED, PASS). No parameter beyond (A, Q, (Z, X, Y)). The instanton survival probability $P_{\text{cand}} = (A/2Q)e^{\{-\pi/A\}} \approx 3.4 \times 10^{-20}$, numerically near the relic-dilution target, is

REJECTED (neither factor derived; selected after seeing the target). The $\sim 10^9$ GeV scale is external context only. The 32/121 normalization is admissible only if μ is fixed by F2 without fitting (F-A19.7-type); fitting μ is the explicit failure mode.

§14. The Closure Program and Non-Claims

Two action-level routes remain on the table: the geometric Z-clock dust of §8 (gates BRL-1–6) and the X-pullback route of v1.0–v1.1 (gates BT-D-1–6 below), retained as a nonlinear completion and cross-check. Each gate is settled by action variation, a Noether/current computation, an external PROVEN theorem, or a finite matrix computation — never by fitting a number.

Table 9. Closure program (Z-Spin Boundary-Tension-to-Dust).

Gate	Requirement	Status
BT-D-1	X-sector slots as action-level comoving maps $X^I : M \rightarrow T^3$.	HYP-strong
BT-D-2	Conserved J^μ from F0/F2 boundary variation as a quantized per-face charge.	OPEN
BT-D-3	Boundary energy $S = -\int \sqrt{(-g)} \mu (-J^2)^{1/2}$, giving $p = 0$, $c_s^2 = 0$, $\rho \propto a^{-3}$.	DERIVED-COND
BT-D-4	$Q_{BT}/Q_{total} = 32/121$ (Theorem A19.2) with no observational fit of μ .	DERIVED-COND
BT-D-5	Adiabatic, passive initial conditions $S_{BT\gamma} = 0$ from a single Z-Spin clock.	OPEN
BT-D-6	Z-anchor as the Jacobian singularity $\det(\partial X) = 0$, linked to the SMBH locator.	HYP-strong
NG1 / trace	Reconcile A_{ZS}/M_{11} via a §7 repair path.	NO-GO+OPEN
A_5 -align	Make $A_5(V_{11})$ compatible with the (2,3,6) blocks and $L_{XY} = 0$.	OPEN
O-A19.6	Resolve the ϵ -Halo overlap so dust and Goldstone halo do not double-count.	OPEN
Boltzmann	Implement boundary-dust perturbations in CLASS/CAMB; reproduce C_ℓ and $P(k)$.	OPEN

Non-claims. **NC-A19.1:** ZS-A19 does not claim the third peak is solved; C1, C2, BT-C and the gate sets are OPEN, and no gate reaches unconditional DERIVED. **NC-A19.2:** does not contradict ZS-A18; a different carrier outside the Goldstone scope, frozen at v1.5. **NC-A19.3:** the $\sim 10^9$ GeV scale and instanton probability are not derivations (anti-numerology). **NC-A19.4:** Z-anchors are locators, not CDM mass carriers. **NC-A19.5:** does not claim one theorem closes both ZS-F23 and the abundance — RETRACTED by A19.NG1. **NC-A19.6:** μ is not derived from ZS-F2; BRL-ZCD removes μ but introduces BRL-1/BRL-3. **NC-A19.7:** $\omega_c = 0.119112$ is a conditional output (on η_B , the boundary lock, and the external BBN coefficient), not an independent measurement; η_B and ω_b are not two successes. **NC-A19.8:** Weinberg constancy is invoked only in the standard single-clock regime. **NC-A19.9:** v2.3 does not claim C1 or C2 are established; both are physical identifications backed by external theorems, not theorems of the corpus.

§15. Conclusion

Every previously blocked route to the Z-Spin third peak assumed dark matter is a collection of objects, and each met an obstruction — over-closure, an underived $\sim 10^9$ GeV scale, or an absent carrier — that is an artifact of that premise. Taking ZS-F2’s definition of CDM as a geometric boundary tension literally, and supplying a conserved pullback / Brown–Kuchař dust, removes the EOS, sound-speed, and over-closure problems at once, with the Goldstone exactly massless and the ZS-A18 NO-GO intact. The fraction 32/121 is a machine-checked algebraic statement: F_{TI} embeds A_5 -equivariantly in the 121-dimensional channel space as a rank-32 projection of canonical trace 32/121, the “32 charge”

disciplined as a projection-class rank (not a spatial-topological charge or a metric tension). Two proven no-gos bound the program: A19.NG1 (BT-C and ZS-F23 Condition C are distinct traces) and A19.NG2 (a bulk scalar-clock rank-lock is absorbed by a multiplier).

It then imports four PROVEN theorems as a conditional reduction (not a closure): they prove the implications but not that the Z-Spin seam satisfies their antecedents. The cross-block claim is re-grounded structurally on the corpus rank-1 β_0 mediation ($\kappa^2 = A/Q = 35/4807 \neq 0$, PROVEN), so $W_{bc} = 0$ would require a block decoupled from the unique mediation mode, not a generic non-cancellation. The restored canonical-branching action S_{branch} , with symplectic-charge conservation $\Theta_c + \Theta_b = \Theta_*$, gives the equal normalization $\varepsilon_c = \varepsilon_b$ an action-level meaning: it is forced once the parent boundary Hamiltonian is single-coefficient linear ($H = \varepsilon_* p$). The named conditions are split into checkable sub-conditions — C1a (physical cube adjacency), C1b (corpus $W_{bc} \neq 0$), C1c (cellular-abelian-BF seam, via Cattaneo–Mnev–Reshetikhin, with the conditional core PROVEN-CONDITIONAL via Eckmann/Lim), C2-norm and C2-branch (equal-Hamiltonian normalization and its action), and adiabatic selection (Weinberg gives an adiabatic mode’s existence, not its occupation). The honest terminus: the physical 32:6 ratio is DERIVED-CONDITIONAL on $C1 \wedge C2$, and the third acoustic peak COMPUTED-CONDITIONAL on $C1 \wedge C2 \wedge$ adiabatic selection $\wedge$ standard Boltzmann transfer, with each sub-condition carrying a proven backbone and an explicit falsification gate. ZS-A19 ends not with a closure but with a precise, falsifiable, partly computable map of exactly what remains — the physical identifications C1c, C2-norm, C2-branch, and adiabatic occupation — with the dust EOS IMPORTED-PROVEN, the integers pre-existing, the ZS-A18 NO-GO frozen, and no new parameter fitted.

v3.0 terminus. v2.4 left C1 and C2 as conditional reductions; v3.0 closes them by the two decisive action-level computations the review named. The Stückelberg→BF dualization of the corpus S10 action, the actual rank-1 β_0 cross-transfer connecting all six baryon nodes to the connected truncated-icosahedron block (eliminating the cube rewiring), and the cellular boundary differential $\Omega\partial\Sigma^0(0\rightarrow 1) = q_Z d_\Gamma$ together raise C1 to DERIVED. The Brown–Kuchař Hamiltonian constraint $P_T = -H_{\text{ZS}}$, the canonical branching with $\Theta_c + \Theta_b = \Theta_*$ and node-uniform normalization $\varepsilon_c = \varepsilon_b$, the vanishing intrinsic entropy transfer, and the BRST-exactness of the relative graph modes raise C2 to DERIVED-CONDITIONAL on a single source. Theorem A19.ZHCS-Closure then makes the 32:6 ratio and $\omega_c = 0.119112$ action-level consequences DERIVED-CONDITIONAL on one sharp cosmological input — single-source reheating with no independent spectator at Σ_* — together with the cellular-discretization exactness of C1-C. This is the strongest honest statement the corpus tools reach: not an unconditional derivation, but a closed conditional theorem whose single residual is a stated, falsifiable cosmological condition rather than a diffuse identification. The dust EOS remains IMPORTED-PROVEN, the integers pre-exist, the ZS-A18 NO-GO is frozen, A19.NG1/NG2 stand, and no new parameter is fitted.

Acknowledgements and Code Availability

This paper consolidates internal Z-Spin Collaboration deep-exploration notes following ZS-A18 v1.5. The corrected polyhedral and graph-spectral results (Appendix C) are reproduced by `zs_a19_verify_v2_1.py` (14 checks) and the cross-transfer genericity (Appendix D) by `zs_a19_ilc_genericity.py` (200,000-sample Monte Carlo), both NumPy/SciPy, no other dependencies. The v2.0 representation-theory, partition, EOS, NG1/NG2, anti-numerology, and CAMB checks are retained as previously archived. External dust, von Neumann embedding, BV-BFV/mQME, cellular BF, combinatorial-Hodge, Brown–Kuchař, and

adiabatic-mode results are cited in full. This work used AI tools (Anthropic Claude) for verification and drafting; the author assumes full responsibility for all content.

Appendix A. Representation-Theory Verification

A₅ table on classes (1A, 2A, 3A, 5A, 5B) with sizes (1, 15, 20, 12, 12); irreducibles 1, 3, 3', 4, 5 (Σ dim² = 60). The 12-vertex permutation character (12, 0, 0, 2, 2) minus trivial gives χ(V₁₁) = (11, -1, -1, 1, 1) = 3 ⊕ 3' ⊕ 5; squaring gives χ(End V₁₁) = (121, 1, 1, 1, 1) with multiplicities (3, 6, 6, 8, 10) and traceless 120 = |I_{-h}|. The pentagon orbit (12, 0, 0, 2, 2) and hexagon orbit (20, 0, 2, 0, 0) sum to χ(F_{-TI}) = (32, 0, 2, 2, 2) = 2(1 ⊕ 3 ⊕ 3' ⊕ 4 ⊕ 5); each multiplicity (2) ≤ that in End(V₁₁), so the rank-32 embedding exists. For A19.NG1, 3m₃ + m₁ + 5m₅ = 11 has the unique solution (1, 3, 1) → weights (3, 3, 5)/11; the minimal amplification realizing 5·(3, 2, 6) is (5, 10, 6) on M₅₅. The pullback pressure p = F - bF_{-b} vanishes for F = -μb.

Appendix B. Dependencies and Cross-References

Table B1. Principal corpus dependencies.

Paper	Role in ZS-A19
ZS-F2	Ω _{-cdm} = F(TI)/Q ² = 32/121; CDM as geometric boundary tension.
ZS-A18 (v1.5)	Massless-Goldstone NO-GO; frozen; defines the scope this carrier lies outside.
ZS-Q11 / ZS-F23	A _{-ZS} = M ₃ ⊕ C ⊕ M ₅ ; canonical trace; Condition C, distinct from BT-C per A19.NG1.
ZS-M9	Truncated-icosahedron 32-dim face module; A ₅ ≅ I structure.
ZS-S10	Vortex Bose/Fermi duality; holonomy (not Dirac) core spinor.
ZS-A1 / ZS-S3	Massless Goldstone ε-Halo and Goldstone theorem (m _{-θ} = 0).
ZS-U3 / ZS-F5	η _{-B} = (6/11) ³⁵ ; η _{-B} →ω _{-b} Independence Warning; used by A19.4.
ZS-F0	GHY term, Robin condition, BV-BFV functor, seam J as boundary operator.
ZS-A15	Broken internal U(1) _{-Z} (supports the abelian structure of condition C1).
ZS-U6	Radiation-era equipartition does not transfer to the matter era (§8.7).
ZS-S1	Z-sector Z ₂ split (one Z ₂ -even β ₀ , one Z ₂ -odd gauge); basis for XQ-1 = 32.
ZS-F10	i-tetration internal time v; ruled out as the dust clock by A19.6.

Appendix C. Corrected Polyhedral / Graph Verification (14/14 PASS)

tO: (V, E, F) = (24, 36, 14), Euler 2; vertex–face incidence rank 12, kernel dim 2 (a uniform square mode and an independent hexagonal mode); the six square faces mutually non-adjacent (0 edges); cube face graph K_{-{2,2,2}} connected (rank 5). tI: (60, 90, 32), Euler 2; incidence rank 31, kernel dim 1; pentagon/hexagon entry ratio -2; (Σv)²/ || v ||² = 4/17; face graph 32 nodes / 90 edges / connected / Laplacian rank 31. Combined 38-node graph (cold ⊕ cube ⊕ one cross-edge) connected with one-dimensional constant Laplacian kernel, returning Q_{-c} : Q_{-b} = 32 : 6.

Appendix D. Conditional-Core Lemma (degree-0 Hodge)

Degree-0 Hodge. For a finite connected graph with N nodes, L₀ = d₀† d₀ has no down-part, so ker L₀ = ker d₀ = span(1_{-N}): a harmonic 0-cochain is constant, with no separate co-closed condition. With P_{-c}, P_{-b} (dims 32, 6; P_{-c} + P_{-b} = I₃₈), Q_{-c} = 32 p★, Q_{-b} = 6 p★, ratio 32 : 6 independent of p★. **Note**

(v2.4): the v2.3 uniform-sector genericity Monte Carlo is SUPERSEDED — a continuous distribution has exact-zero probability 0 by construction, so it could only show that random matrices rarely cancel. The cross-block claim is re-grounded structurally on the corpus β_0 mediation in Appendix F.

Appendix E. Verification Provenance (v2.0 60-check suite → v2.4)

Table E1. Provenance of the verification suite. RETAINED: still valid as stated. SUPERSEDED: invalidated by the corrected substrate. RECOMPUTED: re-derived here with corrected values. NEW: added in v2.1–v2.4.

Check group	Class	Note
A ₅ rep theory, F ₋ TI embedding, $\tau_Q = 32/121$ (Table 1)	RETAINED	Appendix A; unaffected by the ZHCS corrections.
A19.NG1 trace incompatibility; A19.NG2 rank absorption	RETAINED	PROVEN no-gos; intact.
Planck partition 6/32/38/83 over 121 (Table 7)	RETAINED	Face-counting budget; ZS-F2 §11.4.
CAMB third-peak consistency (Table 4)	RETAINED	COMPUTED / OBSERVATIONALLY CONSISTENT, TT only.
tI face graph (32, 90, connected, rank 31); incidence kernel dim 1; $v_p = -2$ v_h ; $(\Sigma v)^2 / \ v\ ^2 = 4/17$	RECOMPUTED	Appendix C; confirms the v2.0 tI values.
tO 6 squares cube-adjacent; “unique tO null mode” cross-support	SUPERSEDED	tO incidence kernel is 2-dimensional; the six squares are mutually non-adjacent (Appendix C). v2.0’s cross-support argument retracted.
38-node graph connected $\Rightarrow$ harmonic charge uniform $\Rightarrow 32:6$	RECOMPUTED	Appendix C; the conditional core, now PROVEN-CONDITIONAL via Eckmann/Lim.
Cross-block $W_{bc} \neq 0$ (uniform-sector Monte Carlo)	SUPERSEDED	Statistically weak (exact-zero probability 0 by construction); replaced by the structural β_0 argument, $\kappa^2 = A/Q$ (Appendix F).
Symplectic branching: $\Theta_c + \Theta_b = \Theta_*$; $\epsilon_c = \epsilon_b$ under $H = \epsilon_* p$	NEW	Appendix F; restores the v2.0 §10.6 branching and makes equal normalization explicit.

Appendix F. Explicit C1/C2 Computation (16th–17th checks)

C1b (cross-transfer structure). The corpus fixes the X–Z–Y mediation as the rank-1 β_0 channel $C_{ZX} = \kappa|z_0\rangle\langle r_X|$, $C_{ZY} = \kappa|r_Y\rangle\langle z_0|$ with $\kappa^2 = A/Q = (35/437)/11 = 35/4807 \approx 7.28 \times 10^{-3} \neq 0$ (PROVEN). The uniform-sector cross-transfer is $W_{bc}(\text{uniform}) = \kappa^2 \langle \text{baryon_uniform} | r_X \rangle \langle r_Y | \text{cold_uniform} \rangle$, vanishing only if a block has zero overlap with the unique β_0 mode z_0 — i.e. a block decoupled from all cross-seam transfer ($L_{XY} \equiv 0$ forces every channel through z_0). This is a structural condition (gate F-A19.7), not a generic non-cancellation; the residual C1b is the explicit nonzero overlaps $\langle z_0 | r_X \rangle$, $\langle z_0 | r_Y \rangle$.

C2 (symplectic branching and equal normalization). With $P_c = \text{diag}(I_{32}, 0)$, $P_b = \text{diag}(0, I_6)$ ($P_c + P_b = I_{38}$), parent charge $p = p_* \cdot 1_{38}$, the branching S_{branch} sets $\Pi_c = P_c p$, $\Pi_b = P_b p$, giving $Q_c = 32 p_*$, $Q_b = 6 p_*$, and $\Theta_c + \Theta_b = \langle (P_c + P_b)p, \delta T_* \rangle = \Theta_*$ (verified identity). Under the single-coefficient parent Hamiltonian $H = \epsilon_* p$, the block energies give $\epsilon_c = H_c/Q_c = \epsilon_b = H_b/Q_b = \epsilon_*$ exactly — equal normalization is forced, not assumed. The constraint surface is parametrized by p

alone, so no independent dust-density integration constant survives; with harmonicity $p = p_* 1_{38}$ (one dof), a perturbation gives $\zeta_c = \zeta_b$. These results are reproduced by `zs_a19_c1c2_compute.py`.

Appendix G. C1 closure computation (18th–20th checks)

C1-A (symbolic). In the parent action S_{par} , integrating out the 3-form H gives $H = f_Z^2 (d\theta - q_Z B_Y)$ and back-substitution returns the Stückelberg kinetic $(f_Z^2/2)(d\theta - q_Z B_Y) \wedge (\cdot)$ (sympy-verified coefficient $f_Z^2/2$); varying θ gives $dH = 0$, $H = dB$, and the dual $S_{\text{dual}} = -(1/2f_Z^2) \int dB \wedge *dB + q_Z \int B \wedge F_Y - (1/2g_Y^2) \int F_Y \wedge *F_Y$. **C1-B.** With $\kappa^2 = A/Q = 35/4807$ and $k_0 = \langle z_0 | K_Z | z_0 \rangle \neq 0$, $W_{bc} = \kappa^2 k_0 |r_b\rangle \langle r_c|$ has all 6×32 entries nonzero ($\min |W_{ij}| > 0$, $\min_i \sum_j |W_{ij}|^2 > 0$). The physical graph (real TI adjacency + six isolated square nodes + actual cross-edges) has one component, Laplacian rank 37, kernel $\text{span}\{1_{38}\}$ — no cube rewiring. **C1-C.** Building d_Γ as the signed incidence of the physical graph, $l\Omega\partial\Sigma^\wedge(0 \rightarrow 1) - q_Z d_\Gamma l = 0$, $\text{rank } d_\Gamma = 37$, $\dim \ker d_\Gamma = 1$. Reproduced by `zs_a19_c1c2_closure.py`.

Appendix H. C2 closure computation (21st–22nd checks)

C2-E. With $P_c = \text{diag}(I_{32}, 0)$, $P_b = \text{diag}(0, I_6)$ and the clock-energy momentum $P_T (= -H_ZS)$, the node-wise normalization $\varepsilon_a = \Pi_{\{T,a\}} / (-\sqrt{h} \rho_a)$ is equal across all 38 nodes ($\varepsilon_c = \varepsilon_b$), and $\Theta_c + \Theta_b = \langle (P_c + P_b)p, \delta T_* \rangle = \Theta_* (P_c + P_b = I_{38})$. **C2-A.** A single parent perturbation gives $\delta Q_c/Q_c = \delta Q_b/Q_b$, so $S_{cb} = 0$; on the uniform-density slice the intrinsic transfers vanish, giving $S_{c\gamma} = 0$, conditional on a single source. **Relative modes.** L_Γ has $\lambda_1 = 0$ (uniform) and $\lambda_2, \dots, \lambda_{38} > 0$; in the topological BF these 37 modes are BRST-exact, so $H^0_{\text{phys}}(\Gamma_m) = \text{span}\{1_{38}\}$, $\dim 1$. **Consequence.** $\rho_c/\rho_b = 32/6$, $\omega_c = (32/6) \omega_b = 0.119112$. Reproduced by `zs_a19_c1c2_closure.py`.

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Version History

v1.0 (June 2026): Initial release. Geometric-dust ontology, equivariant boundary-rank–trace theorem, two NO-GO characterizations of the 32-charge; Condition BT-C and O-A19.2–O-A19.5 registered.

v1.1 (June 2026): Added Theorem A19.NG1; RETRACTED the v1.0 “one theorem closes both”; demoted A19.1 to IMPORTED-PROVEN; corrected pullback pressure $p = F - bF_b$; elevated O-A19.5; P_BT a candidate; added ε -Halo conflict O-A19.6.

v1.2 (June 2026): Added §8 BRL-ZCD (XQ–1 seam projector, constrained Z-clock dust, baryon-anchored lock); Theorem A19.4 (h-free $\omega_c = 0.119112$, ratio 16/3); AN-A19.2.

v1.3 (June 2026): Gate-testing revision (CAMB peak, S_3 uniqueness, $\delta_c = \delta_b$, i-tetration clock); several conclusions corrected in v1.4.

v1.4 (June 2026): Added Theorem A19.NG2 (scalar-clock rank-absorption); RETRACTED the v1.3 single-emergence-node claim; downgraded A19.5, BRL-5, BRL-1/2/4; honest gate inventory G1–G8.

v1.5 (June 2026): Added §10 ZHCS boundary-charge program; reduced G1–G8 to four theorems ZHCS-1–4; rank-ratio PROVEN-CONDITIONAL on $ZHCS-1 \wedge ZHCS-2$; rank-6 projector by direct sum.

v2.0 (June 2026): Deep-exploration closure: claimed ZHCS-1 DERIVED (38-node graph connected) and ZHCS-2 DERIVED-CONDITIONAL, raising 32:6 to DERIVED-CONDITIONAL on a single operator identification; added the ZHCS-1 polyhedral computation (60-check suite).

v2.1 (June 2026): Status correction. Independent recomputation: truncated-octahedron incidence kernel dim 2, six squares mutually non-adjacent, mQME restated; demoted ZHCS-1 $\rightarrow$ DERIVED-CONDITIONAL, ZHCS-2 $\rightarrow$ OPEN/HYPOTHESIS-strong; 32:6 = PROVEN-CONDITIONAL on $ZHCS-1 \wedge 2 \wedge 3$; conditional core lifted via Eckmann/Lim; corrected title 32/121 $\rightarrow$ 32:6; separated third-peak from abundance ratio.

v2.2 (June 2026): External-theorem closure program. ZHCS-1 repaired via uniform-sector Schur genericity (codim-1 vanishing, 200k-sample MC); ZHCS-2 collapsed to condition C1 (cellular abelian BF, CMR), co-closedness dissolved at degree 0; ZHCS-3 and ZHCS-4 unified under condition C2 (single Brown–Kuchař Z-clock) via Brown–Kuchař + Weinberg; physical 32:6 = DERIVED-CONDITIONAL on $C1 \wedge C2$; third peak = DERIVED-CONDITIONAL on C2.

v2.3 (June 2026): Full integration. Restores all v2.0 content omitted by v2.1/v2.2 (geometric-dust ontology §3–§4, equivariant rank–trace §5, the 32-charge NO-GOs §6, BT-C/A19.NG1 §7, BRL-ZCD §8, the G1–G8 gate testing §9, O-gates §11, the Planck partition §12, BTD closure program §14, Appendices A–B, references [1]–[21], and the full

version history) and integrates them coherently with the v2.1 substrate corrections and the v2.2 external-theorem consolidation, applying the corrected ZHCS statuses uniformly throughout (Table 6, §12, §13). Net result: physical 32:6 DERIVED-CONDITIONAL on $C1 \wedge C2$; third peak DERIVED-CONDITIONAL on $C2$; no gate unconditionally DERIVED. Four external PROVEN theorems imported ([22]–[27]). ZS-A18 frozen; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED; no new fitted parameter.

v2.4 (June 2026): Conditional-reduction revision answering a review of v2.3. (1) Restored the canonical-branching action S_{branch} and symplectic-charge conservation $\Theta_c + \Theta_b = \Theta_*$ (dropped in v2.3), giving the equal normalization $\varepsilon_c = \varepsilon_b$ an action-level basis; (2) re-grounded ZHCS-1 cross-support structurally on the corpus rank-1 β_0 mediation ($\kappa^2 = A/Q = 35/4807 \neq 0$, PROVEN), retiring the statistically weak uniform-sector Monte Carlo; (3) split $C1$ into $C1a$ (physical cube adjacency), $C1b$ (corpus $W_{bc} \neq 0$), $C1c$ (cellular-abelian-BF seam), and $C2$ into $C2\text{-norm}$ (single-coefficient parent Hamiltonian $H = \varepsilon_* p$) and $C2\text{-branch}$ (S_{branch} from ZS-F0); (4) made the equal-Hamiltonian-normalization condition explicit and proved $\varepsilon_c = \varepsilon_b$ follows from it (Appendix F); (5) corrected ZHCS-4 to PROVEN-CONDITIONAL on $C2 + \text{no-independent-entropy} / \text{adiabatic selection}$ (Weinberg gives existence, not occupation), narrowing the co-moving identification to the branching hypersurface to remove the conflict with baryon acoustic physics; (6) corrected the third peak to COMPUTED-CONDITIONAL on $C1 \wedge C2 \wedge \text{adiabatic selection} \wedge \text{standard transfer}$ (not $C2$ alone); (7) reclassified “external-theorem closure” as “external-theorem conditional reduction”; (8) added Appendix E (verification provenance: RETAINED / SUPERSEDED / RECOMPUTED / NEW) and Appendix F (explicit $C1/C2$ computation). No gate reaches unconditional DERIVED; $C1c$, $C2\text{-norm}$, $C2\text{-branch}$, and adiabatic occupation remain physical identifications. ZS-A18 frozen; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED; no new fitted parameter.

v3.0 (June 2026): Closure revision. Carried out the two decisive action-level computations of the v2.4 review. (C1) Dualized the corpus $S10$ Stückelberg $U(1)$ to a 2-form–1-form BF coupling $q_Z \int B \wedge F_Y$ ($C1\text{-A}$, symbolic); computed the actual rank-1 β_0 cross-transfer $W_{bc} = \kappa^2 k_0 |r_b\rangle\langle r_c|$ with all 6×32 entries nonzero, giving a connected physical 38-node graph (TI adjacency + six isolated squares + actual cross-edges, rank $L_\Gamma = 37$) with the cube rewiring ELIMINATED ($C1\text{-B}$); and showed the boundary BFV differential $\Omega \partial \Sigma^\wedge(0 \rightarrow 1) = q_Z d_\Gamma$ ($C1\text{-C}$) — raising $C1$ to DERIVED. (C2) Performed the Brown–Kuchař ADM reduction giving $P_T = -H_{\text{ZS}}$, the canonical branching with $\Theta_c + \Theta_b = \Theta_*$ and node-uniform $\varepsilon_c = \varepsilon_b$ without separate assumption ($C2\text{-E}$), vanishing intrinsic entropy transfer $S_{cb} = S_{cy} = 0$ ($C2\text{-A}$), post-branching separation ($C2\text{-D}$), and BRST-exactness of the 37 relative modes ($H^0_{\text{phys}} = \text{span}\{1_{38}\}$) — raising $C2$ to DERIVED-CONDITIONAL on a single source. Added Theorem A19.ZHCS-Closure (PROVEN-CONDITIONAL on $C1 \wedge C2$): $\rho_c/\rho_b = 32/6$, $w_c = c_{s^2} = 0$, $S_{cb} = S_{cy} = 0$, standard passive CDM transfer, $\omega_c = 0.119112$ — all DERIVED-CONDITIONAL on a single source. Added Appendices G–H ($C1/C2$ closure computations) and the script `zs_a19_c1c2_closure.py`. All v2.4 content retained; the only residuals are the single-source condition ($C2\text{-A}$) and the cellular-discretization exactness ($C1\text{-C}$). ZS-A18 frozen; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED; no new fitted parameter.